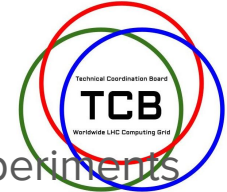


WLCG Technical Evolution: Preparing for HL-LHC with a Community-Driven Roadmap

James Letts (UCSD) and Alessandro Di Girolamo (CERN)
on behalf of the WLCG Technical Coordination Board ([TCB](#))
[CHEP26](#) - May 26, 2026

WLCG Technical Evolution and Coordination

WLCG: Experiments, Sites, and People

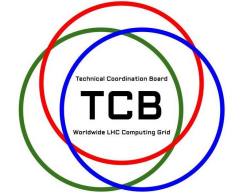


- The primary goal of WLCG is to **enable the scientific programs** of the LHC experiments by providing the required computing infrastructures and the accompanying services at the necessary scales.
- The **physical infrastructure is provided by some 160 sites** connected by high-speed research networks all around the world.
- The people who work at these facilities are often the **same people planning and executing the evolution** of the infrastructure, middleware, and software that makes up the “WLCG”, **therefore very much a “bottom-up” flow of ideas (and “ownership”!)**



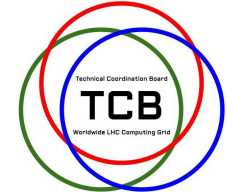
Many more details in Simone's plenary [presentation](#) yesterday.

WLCG Technical Evolution



- WLCG [Technical Coordination](#) was established in Oct. 2024 to **coordinate this evolution, identify “gaps”** in scale or technology, and **outline a strategic technical roadmap** to bridge them.
- **There are several fora** where different aspects of the technical evolution are discussed, some in WLCG (e.g., the [WLCG Open Technical Forum](#), operations coordination, [DOMA project](#), various [Task forces](#) and [working groups](#)), others not (e.g., HEPiX Working Groups, IRIS-HEP, many others).
- Our aim is to show you that **we are building a concrete, credible plan** to bring WLCG sites and services towards a shared goal (which will demonstrate to our reviewers e.g., LHCC, that we will be ready for Run 4).

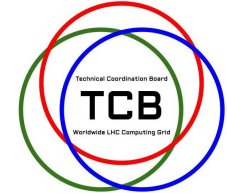
WLCG Technical Roadmap



The Chapters of the WLCG Technical Roadmap have been **organized thematically**, though there are many inter-relationships between the topics, and other possible ways to organize the flow of the document.

1. Experimental Computing Requirements (setting the goals)
 2. Facility Evolution (infrastructure providers)
 3. Data Management
 4. Network Infrastructure and Management (“middleware”)
 5. Workflow Management
 6. Security and Authentication & Authorization Infrastructure
 7. Services
 8. New Architectures and Infrastructures (still many unknowns in the requirements)
 9. Sustainability
- We will be **explicit about where we have an agreed-upon plan and where we don’t** (yet).
 - We don’t need a lot of text (a few pages per topic). We need good text.

WLCG Technical Roadmap



What is a Roadmap? It's a high-level **outline of a journey**, not just a destination.

Like any journey it has:

- Milestones to guide the way
- Decision points - or forks in the road
- Can be fraught with risks that could have serious impacts
- And strategies to mitigate those risks

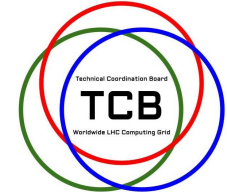


We will summarize the plan in each of the chapters in the rest of this presentation.

- In many cases the plans are quite advanced and agreed-upon within the community.
- In other cases, the milestones are tentative, the path forward less certain, so we'll point out explicitly where that is the case.
- Many details are in the backup slides!

Setting the Scene: Experiments' Requirements

Chapter 1: Experiment Requirements (Timelines)



The WLCG infrastructure and services that power LHC data processing, simulation, and analysis **need to scale up** to HL-LHC requirements and adapt to **added resource complexity**.

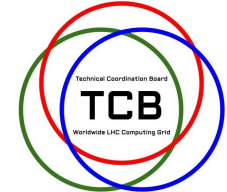
Experiment **timelines** (*effectively milestones*) are equally important:

- The computing resource request process for 2031 (the first *full year* of Run 4) **begins in mid-2029**, reviewed/scrutinized, and then approved at the April 2030 RRB meeting.
- The WLCG infrastructure providers (sites) match the needs with pledges by Sep. 2030 and deploy resources by April 2031.

Networking (which ties the sites together) has even longer lead times for design and deployment - the requirements need to be understood already now.

All computing, software/middleware, analysis, and data challenges need to take place between now and 2029 at the latest.

Chapter 1: Experiment Requirements (Scale)



Example from the recent CMS Software and Computing CDR, [CERN-LHCC-2026-003](#)



- Run 4 CPU and disk resource requirements are *roughly double* Run 3, and **double again** for Run 5 (and tape $\sim 3x$). **Depends on R&D success!**
- Outside the gray “flat-budget” envelopes (see [next slide](#)), which were **determined before the recent AI build-out**.
- Driven by increases in the data volume, detector upgrades, **mitigated by R&D!**

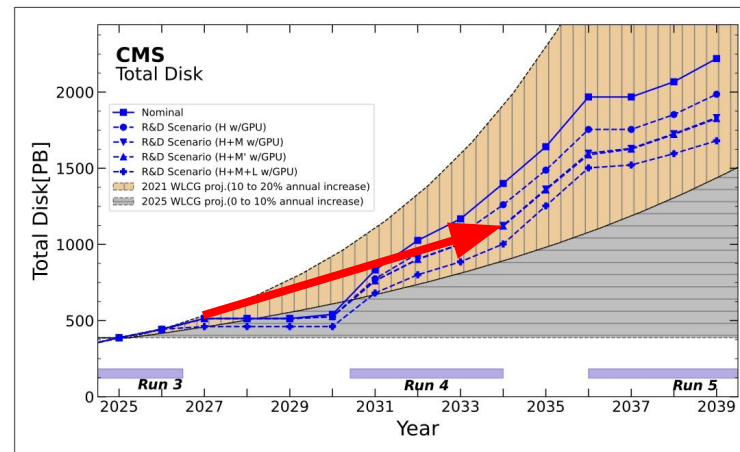


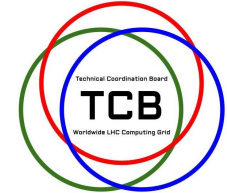
Figure 12: Estimated total disk storage needs annually through the middle of HL-LHC Run 5. Blue curves correspond to annual projections of needs for the baseline scenario and separately each of the R&D scenarios. Two bands (grey, with horizontal hatching, and orange-brown, with vertical hatching) correspond to the current and previous WLCG guidance on anticipated resource growth.

We expect the ATLAS TDR later this year - similar magnitude of increases? (See also: [ATLAS CDR](#), 2022)

ALICE and LHCb Run-4 resource requirements increasing adiabatically from Run 3 levels. More evolution for Run 5.

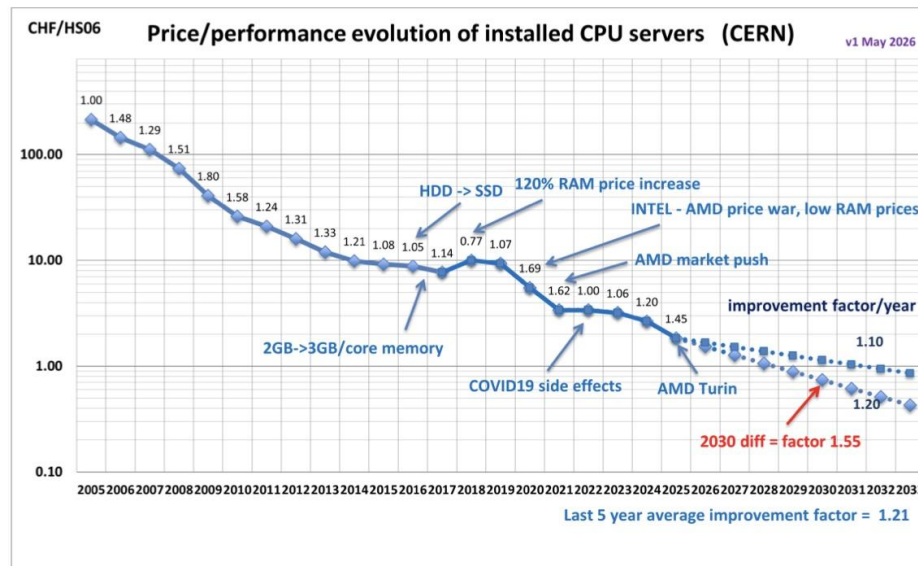
N.B. Requirements for heterogeneous Architectures (GPUs) are discussed later in [Chapter 8](#).

Major (Realized) Risk: Hardware Cost Evolution



Multi-year disruptions from the long-term technology cost trends are nothing new.

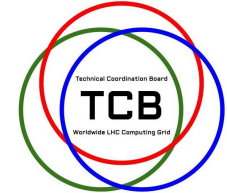
- HEPiX [Technology](#) and [Benchmarking](#) Working Groups track long term trends.
- “Flat budget” y/y cost evolution [agreed upon last year](#) upended by the “AI boom”. (5%-10% y/y cost *improvement*)
- Current **strategy** (agreed with the LHCC in the RRB) is to **monitor the market** and re-discuss if it doesn’t rectify within a reasonable amount of time (which could be a few years).



See [yesterday's presentation](#) by A. Sciaba and a recent [talk](#) at the HEPiX Spring 2026 Workshop.

Technical Evolution of Facilities, Middleware, and Services

Chapter 2: Facility Evolution



We have had two OTF meetings co-hosted with our partners at HEPiX at their Spring Workshops, where we discussed facility evolution plans from various WLCG Federations:

- At [OTF#4](#) (Apr. 2025): Italy, Germany, U.S. ATLAS, and others
- At [OTF#10](#) (Apr. 2026): U.S. CMS and China (IHEP)

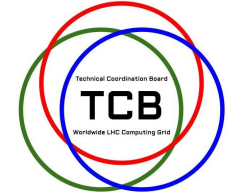
*Germany (see [next slide](#)) and U.S. CMS have quite similar facility evolution plans, **concentrating storage** at national laboratories and **dispersing processing** (including GPUs, analysis infrastructures) across HPCs or university (Tier-2) sites. Others more conservative.*

Milestones: Leverage WLCG DOMA mini-challenges on tape performance, storage capacity vs. throughput, operational effort needs (all extensively discussed at the HEPiX Spring 2026 Workshop/OTF#10) to test the feasibility of evolving infrastructure configurations. (See the [backup slides](#) for more details of the plan). See CHEP26 presentations related to mini-challenges [here](#) and [here](#).

Chapter Facilitators: A. Dewhurst, J. Flix, A. Melo. A. de Salvo

WLCG Technical Roadmap - CHEP26 - May 26, 2026

Chapter 2: Facility Evolution

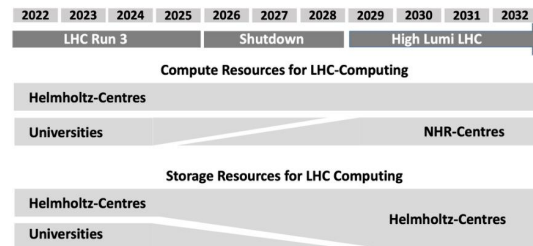


Example: German facility **evolution** plan (HEPiX Spring Workshop 2025):

Transformation towards HL-LHC

No new hardware @ universities in Germany

- ▶ Gradual transition of university-based Tier-2 resources:
 - ▶ **CPU** to 3 selected National High Performance Computing (NHR) centres
 - ▶ **Mass storage** to Helmholtz-Centres (HGF)
 - ▶ Storage at HGF: DESY-HH and KIT will add/increase Tier-2 disk storage
- ▶ Local ATLAS/CMS groups keep supervising the NHR resources (funding for personnel) granted
- ▶ Motivation:
 - ▶ higher energy efficiency
 - ▶ better QoS at Helmholtz centres



Strategy paper of the German Committee for Elementary for Particle Physics from 2022:

https://www.ketweb.de/sites/site_ketweb/content/e199639/e312771/KET-Computing-Strategie-HL-LHC-final.pdf

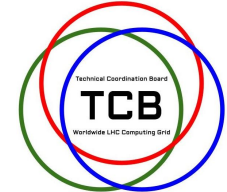
universität freiburg

Albert-Ludwigs-Universität Freiburg | Dr. Michael Böhler | April 1st 2025

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See a related CHEP26 [presentation](#) tomorrow.

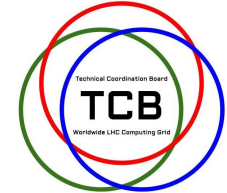
Chapter 3: Data Management - DOMA



- [DOMA](#) (Data Organization, Management and Access) WLCG R&D project.
 - Address challenges of managing and accessing the enormous [amounts of data](#) expected during the High-Luminosity LHC (HL-LHC) era.
 - Focuses on developing, prototyping, and **testing novel approaches to distributed storage, data transfers, protocols, and networking** to meet those challenges.
- Key aspects of the well-defined plan:
 - **Data challenges** like DC24 (25% HL-LHC scales, where we learned a lot) and next DC27 (50% HL-LHC, last week of Feb. and first week of March 2027).
 - **Mini-data challenges** testing throughput capacity and technology capabilities of infrastructure. See more details of the overall plan in the [backup slides](#).
 - Tools and technologies: Rucio for Data Management, FTS (File Transfer Services, see [next slide](#)), XRootD (see the [backup slides](#), including for [XRootD monitoring](#)), [HammerCloud](#) for site and workload testing, and network monitoring with perfSONAR.

Chapter Facilitators: K. Ellis, J. Elmsheuser (DOMA Coordinators), with contributions from the FTS and XRootD Developers.

Chapter 3: Data Management - FTS



FTS4 will carry us through the beginning of HL-LHC, bridging the gaps seen in DC24:

- Essential for DC27 success.
- One Goal: Correct scheduling decisions under heavy load.
- Recent Experiment scale tests have helped accelerate the development cycle.
- DC27: internal release CERN-only ready in late 2026.
- General release aligned with RH10/Alma10 with full support for sites after DC27.



Main FTS goal for DC27

Why FTS3 cannot do DC27	What FTS4 delivers for DC27
1 Wrong decisions under load Starvation, priority failures, capacity limits not respected: design flaws, not bugs	Correct scheduling decisions under load – No starvation: critical for tape recalls – Activity-shares for guaranteed file-transfer concurrency – Priorities for rare queue jumping – Tape factor mode: explicitly and simply favour tape – Data-challenge mode: correct the mistakes of DC24
2 Feature development blocked Underlying complexity has blocked experiment requirements for years	
3 Non-deterministic behaviour Stochastic scheduler — decisions cannot be explained or validated at DC27 scale	
4 Database hits a hard ceiling Shared scheduler and monitoring cannot scale to 20M queued requests	

CERN only for DC27. General release aligned with ALMA10 — external sites get a planned rollout with full support, not a last-minute deployment.

WLCG/DOMA General Meeting 4

[DOMA General Meeting](#), April 29, 2026.

See also a CHEP26 presentation on FTS4 development [here](#).

Here again there is a well-defined plan. See a [backup slide](#) for details and timelines.

Chapter 4: Network Management



Core Challenge: Sub-standard network capacity directly translates into higher data transfer latency, workflow delays, and operational overhead, adversely impacting time-to-science.

WAN Connectivity Targets (Gbps)

Site Type	2024	2027	2029	2030-32 (HL-LHC Ramp-up)	Full HL-LHC Rate (2033+)
Tier-2(small)	10 Gb	20 Gb	20-40 Gb	20-40 Gb	40 Gb+
Tier-2 (normal)	40 Gb	100 Gb	100-200 Gb	100-400 Gb	200-400 Gb
Tier-2(large)	100 Gb	200 Gb	400 Gb	400-800 Gb	800-1200 Gb
Tier-1	200 Gb	400 Gb	400-800 Gb	800 Gb+	800 Gb+ (multi links)

Key Capabilities Beyond Bandwidth

- **Monitoring:** perfSONAR (some ≥ 100 Gbps), SciTags flow marking
- **IPv6-Only:** Phase out dual-stack by 2030
- **Optimization:** BBR congestion control, jumbo frames, QoS
- **Orchestration:** SENSE/Rucio, SDN for dynamic path management
- **Automation:** Analytics for problem detection & remediation

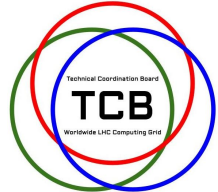
Timeline & Governance

- **2026:** IPv6-only transition starts (LHCOPN), 50% perfSONAR rollout
- **2027:** Jumbo frames in widespread use, security recommendations
- **2028-2032:** Incremental WAN upgrades (DC27/DC29 data challenges)
- **Coordination:** LHCONE, HEPiX, WLCG DOMA, RNTWG

See related CHEP26 presentations on IPv6 [here](#) and [here](#), [packet marking](#) with SciTags, and others.

Chapter Facilitators: S. McKee and E. Martelli

Chapter 5: Workflow Management



Strategic context: While all experiments share a common WLCG infrastructure, **each maintains their own Workflow Management (WM) systems** (and software stacks).

However, all experiments face common challenges:

- More **heterogeneous resource** landscape: co-scheduling CPU, GPU, memory, etc. for an *efficient* use of these fluid resource pools.
- More **heterogeneous backends** as well: traditional Grid, HPC, Cloud, and container-native platforms.
- **New classes of workflows**: ML training, interactive analysis, etc.

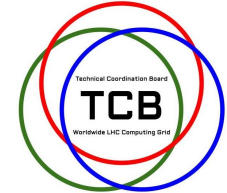
Takeaway: Workflow Management systems must evolve toward capability-aware orchestration of dynamic and heterogeneous resource pools.

We have discussed expanding the [mandate](#) of the [WLCG Job allocation and Handling WG](#) to help coordinate work in this area. **The plan is still evolving.**

Chapter Facilitators: M. Mascheroni, recently joined by L. Betev, A. Perez-Calero, and A. de Salvo



Chapter 6: Services - Accounting

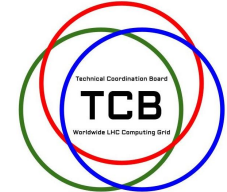


- Accounting: performance monitoring, MoU tracking, resource utilisation:
 - Current limitations: outdated (20 years old) system not fit for heterogeneous, HL-LHC-scale resources. Validation very complex.
- New strategy:
 - Site-level aggregation (monthly summaries)
 - Simplified central services (store, validate, present)
 - Unified, modern architecture
- Key components: AUDITOR (sites) → PEAR (central) ← WSSA (storage)
- Strategic goals:
 - Simpler data flows, better data quality
 - Enable flexibility, support for new (incl. environmental) metrics
 - End-to-end automated validation at the site and central level
- Collaborative effort: WLCG Ops, STFC, AUDITOR, EGI, and supported by EU ENSURE.

See a related CHEP [presentation](#) Thursday.

Chapter Facilitators: J. Andreeva, P. Paparrigopoulos

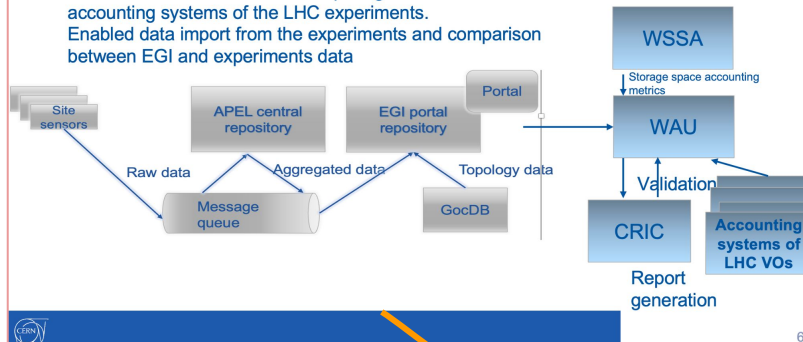
Chapter 6: Services - Accounting



WLCG Operations presented the planned evolution for the Accounting System at [OTF#9](#) on March 25, 2026:

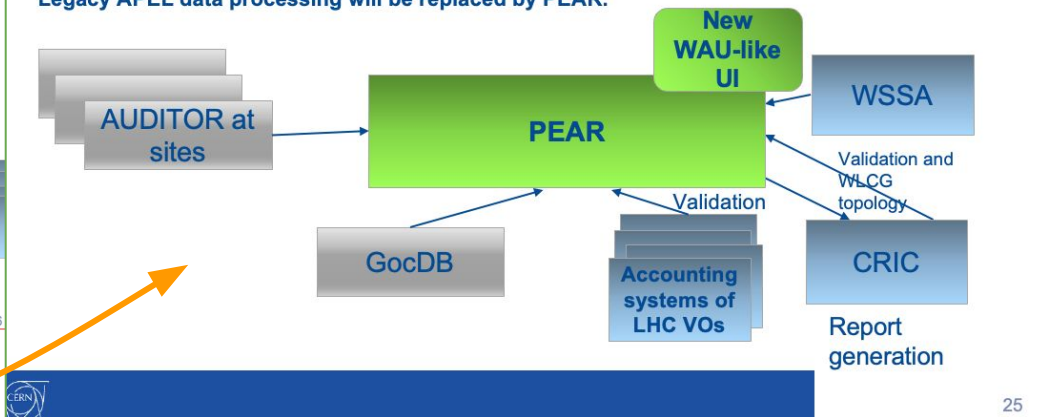
Accounting. A bit of history (3)

Another level of validation is comparing data with the accounting systems of the LHC experiments. Enabled data import from the experiments and comparison between EGI and experiments data



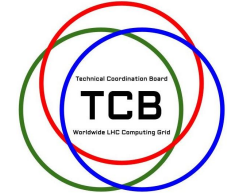
Where we would like to get

In the longer term, we plan to deploy AUDITOR across the majority of WLCG sites. We are also considering replacing the current reporting mechanism with a pull-based model, eliminating the use of message queues. Legacy APEL data processing will be replaced by PEAR.



Again, there is a plan for technical evolution. More details can be found in a [backup slide](#).

Chapter 6: Services - CRIC



- **CRIC is the topology and configuration system** of WLCG, ATLAS, and CMS
 - Constantly evolving together with the WLCG and experiments needs
 - Production-grade service with 3 instances hosted at CERN IT
- **Recently modernised:**
 - Full redesign & Python 3 migration (not just a port) for WLCG and ATLAS instances
 - ATLAS CRIC in production (Mar. 2025); WLCG followed (Autumn 2025)
 - Zero disruption to operations throughout migration
- **Evolving for HL-LHC:**
 - Changes to support Storage Space accounting
 - **GPU topology:** providing information, based on the needs of the community
 - CMS Python 3 migration planned for LS3
 - A prototype of hosting FTS configuration in CRIC (waiting for FTS4)
 - MCP server for AI agent integration under exploration
- See a related CHEP [presentation](#).



Actionable plan in place: See a [backup slide](#) for more details.

Security and Authentication

Chapter 7: Security and AAI (Token Transition)

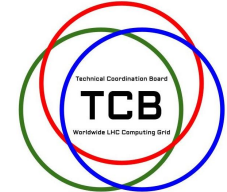


- JSON Web Tokens are an **industry standard** and can be **tuned** along 3 axes:
 - **Lifetime**
 - **Scope** – what precise activities are authorized by a given token
 - **Audience** – which services should accept a given token
- Users do **not need to know** about tokens – tools acquire them as needed.
 - Users will “log into” the Grid e.g., once per day – details to be decided, and may differ by VO
- Grid middleware has to decide **which token to acquire** for which operation(s).
 - Instead of just using ubiquitous VOMS proxies for everything
- Since 2020, the use of tokens has **steadily increased** in production workflows.
 - Pilot job submission to CEs
 - FTS disk-to-disk transfers; tape operations are to follow later this year
- We have **good ideas** for the remaining areas (TBD):
 - Tokens used by grid jobs
 - User workflows

**Plan (mostly) in place:
See a [backup slide](#) for
more details.**

Chapter Facilitators: J.C. Luna (Security) and M. Litmaath (Token Transition Task Force, WLCG Operations)

Chapter 7: Security



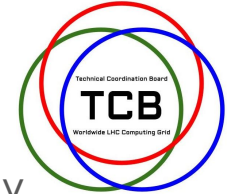
Work is proceeding along four themes:

- **Review of WLCG policies**
 - Identify gaps, inconsistencies, and outdated assumptions (or outdated policies).
 - Harmonization of policy workflow (where to actually store them / format / copies / etc.)
 - Adaptation of the MoU to be more clear and strengthen security importance.
 - Self-assessment checklist for important subset -> maturity indicator
- **Improve containment workflows for major security incidents**
 - Ensure/validate/clarify ability to act, existing workflows and know-how: users, sites, VOs
 - Sites can self declare offline
 - VOs should be able to suspend/remove sites from use until incident contained.
 - Promote documented playbooks + coordination mechanisms
- **Identify skill / improvement gaps**
 - Recommendations/blueprints (e.g., [SOC working group](#)), and computer forensics training.
 - What is missing / needed?
- **Incident transparency and news (exploring options)**
 - Not all incidents are reported... how can we ensure everyone learns from these?

**Tentative plan in place:
See [backup slides](#) for
more details.**

New Architectures and Infrastructures

Experiment GPU Requirements



At the [WLCG Workshop on Heterogeneous Architectures](#) in Dec. 2025, and subsequently, we learned about experiment's capabilities for offloading processing to GPUs (on WLCG resources, Filter Farms, HPCs ...):

- ALICE: 50-60% of the *offline* reconstruction (not simulation) offloadable to GPUs now, working towards 60-80%.
- CMS: 15%-40% of total offline processing needs, depending on R&D progress.
- LHCb: 83% of Grid CPU usage is for full simulation, which now can run on AdePt with 2x speed-up.
- ATLAS: Working towards 15% of the total offline needs (full simulation only).



If one NVDA T4 can offload 0.7kHS23...

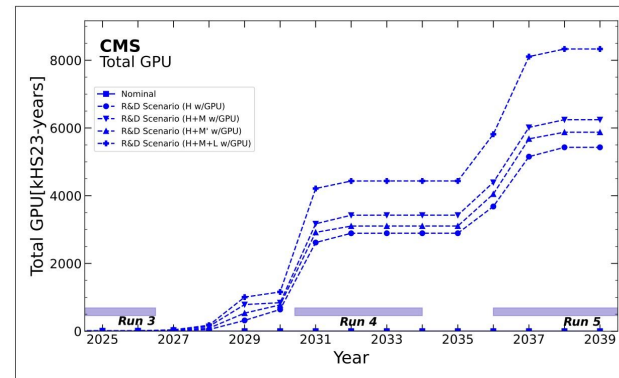
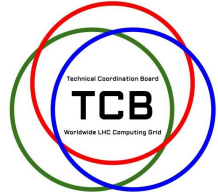


Figure 14: Estimated total GPU processing needs annually through HL-LHC Run 5. Need is expressed in terms of HS23 that has been offloaded. Blue curves correspond to annual projections of needs for the baseline scenario and separately for each of the R&D scenarios. Section 11.3.1 describes current estimates for the conversion of HS23 units into the actual needed GPUs.

*Quite varied - no experiment is ready to **request GPUs** at this time (not the case with ARM).*

Chapter 8: New Architectures and Infrastructures



In summary, the four main macro areas we see for GPUs are:

- Online and Offline **Experiments' Software and Workflows** (see [previous slide](#))
- **Benchmarking** (HEPScore4GPU): What is the HS23 equivalent of CPU-GPU systems? This is more complicated than benchmarking CPU because it will evolve over time as workflows make better use of the GPU. A prototype with existing experiment workflows is already available (see [next slide](#)).
- **Information System/Accounting**: How is the information captured and reported? How dynamic is it? How detailed is it? This is more complex than with CPU. 
- **Provisioning** (WLCG facilities, Cloud, HPC, etc): How do the resources need to be presented in order for WLCG workloads to exploit? 

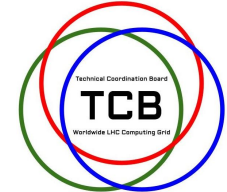
As WLCG, our role now is to be *technically* ready to be able to take advantage of opportunities (or obligations) to run on GPUs as they arise, also for analysis infrastructure on pledged resources.

- Total-cost-of-ownership and pledging are out-of-scope of this Chapter. A cost/benefit analysis may influence our resource requests: Value in having flexible software stacks.

Slide adapted from a [Summary Talk](#) on the WLCG Workshop given at the HEPiX Spring 2026 Workshop.

Chapter Facilitators: O. Smirnova, D. Benjamin, I. Osborne

GPU Benchmarking Milestones & Dependencies

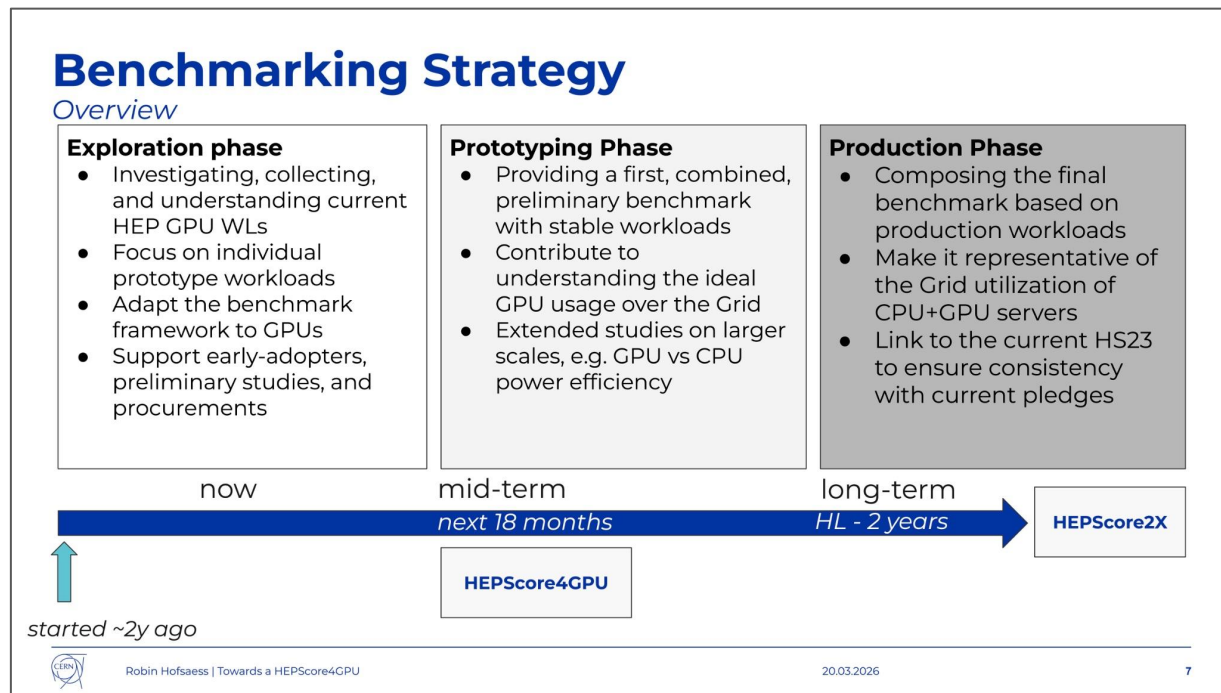


Information systems, accounting, and resource provisioning roadmaps for GPUs can sit in other chapters.

For benchmarking, see the [next plenary presentation](#) by R. Hofsaess.

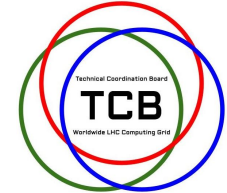
The timeline depends on receiving experiment workflows ready to be included in the suite.

Taken from a [recent presentation](#) by R. Hofsaess & D. Giordano



Timeline in sync with resource request process (HL - 2 years).

Chapter 9: Sustainability



Sustainability issues will be addressed the various chapters:

- Operational sustainability: risk of declining effort under heavier load.
- Technical sustainability: legacy software (major AI security target!)
- Environmental sustainability / total cost of ownership

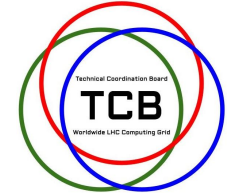
Chapter 9 will summarize these in a brief, coherent way and propose some mitigations in harmony with the findings of the [WLCG Environmental Sustainability Forum](#). See a related CHEP26 [presentation](#). There is also a HEPiX track on sustainability.

Note on AI: Operational intelligence at sites, AI in the evolution of security threats and incident response, intelligent networking, are **just some examples** of *new* and rapidly evolving areas, and may also require changes to how we approach **workforce development** (job descriptions).

We are here at CHEP - We want to hear from you about how we can turn promising ideas into shared, deployable, and trustable solutions!

Conclusions

Conclusions



We are working towards a solid plan for WLCG technical evolution over the next few years on the road to Run 4!

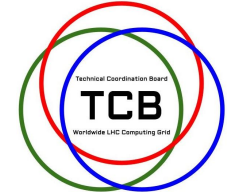
- The first version of the technical roadmap will be reviewed by the LHCC later this year.
- Most of the areas have detailed timelines and plans that give confidence to this important process of WLCG technical evolution.
- Some areas need further work, discussion, refinement, and consensus-building over the next few months e.g.,
 - Many aspects of requesting, provisioning, monitoring, and accounting for New Architectures and Infrastructures,
 - Workflow Management,
 - Tokens for user jobs

We need you, the community, to participate in this process.

There are interesting challenges ahead!



Acknowledgements



We thank all of our colleagues for their contributions to the WLCG Technical Roadmap, Open Technical Forum Meetings, and the Technical Collaboration Board, including our partners in HEPiX, throughout the WLCG, in the experiments, software and middleware development teams, and at the facilities ... a list too long to write here, but colleagues without whose work none of this would have been possible.

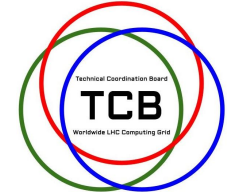
Finally we would like to thank Thailand, Chulalongkorn University, P. Srimanobhas, and all of the organizers (LOC, PC, IAC...) of CHEP26 for making this presentation possible.

ขอบคุณครับ



Backup Slides (for information)

*Including detailed milestone tables
referenced during the talk.*

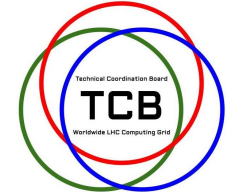


WLCG Technical Evolution: Preparing for HL-LHC with a Community-Driven Roadmap

Authors: The WLCG Technical Coordination Board: J. Andreeva, D. Benjamin, L. Betev, C. Bozzi, D. Britton, I. Chakaberia, B. Couturier, A. De Salvo, A. Dewhurst, A. Di Girolamo, K. Ellis, J. Elmsheuser, J. Flix, J. C. Luna, M. Litmaath, S. Lammel, J. Letts, M. Mascheroni, S. McKee, A. Melo, E. Moyse, I. Osborne, P. Paparrigopoulos, S. Piano, E. Sexton-Kennedy, O. Smirnova, D. South, N. Srimanobhas, F. Stagni, and J. van Eldik.

With the end of Run 3 of the LHC approaching, the Worldwide LHC Computing Grid (WLCG) is entering an important transition toward the HL-LHC era. To meet the substantial increase in data volume, computational requirements, and resource heterogeneity, while preserving reliability, sustainability, and community cohesion, we have launched the development of the WLCG Technical Roadmap 2026-2030. The Roadmap is a consensus-driven, community-owned actionable plan that identifies key gaps and defines concrete milestones across the infrastructure and services landscape. It is organized into nine chapters each outlining a major area of evolution. Planning for WLCG facilities evolution, data management, networking, workflow management, authorization and authentication (tokens), security, WLCG services, heterogeneous resources and new infrastructures (for example, leveraging GPUs) are presented, taking into account that experimental requirements are still evolving. By presenting this roadmap at CHEP, we seek to engage the broader HEP community to gather feedback, foster collaboration, and build collective awareness. As one of the pillars of LHC physics and a cornerstone for data-intensive science even beyond the LHC, WLCG's future relies on coordinated innovation and active community participation.

More WLCG-TR-Related Presentations at CHEP26

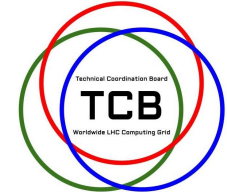


- 170. [Global Grid User Support \(GGUS\) for WLCG and EGI: From Legacy to Next-Generation Helpdesk](#), Pavel Weber (KIT)
- 271. [A workflow based approach to risk assessment and analysis arising from token based authentication within the WLCG](#) Tom Dack (STFC UKRI)
- 138. [From CPU-Centric to Accelerator-Aware: Operational Deployment of MIG and vGPU Partitioning in WLCG](#), Dr Brij Kishor Jashal (RAL)
- 267. [Advancing Workflow Validation at Scale: A Modern and Containerized HammerCloud Architecture for the WLCG](#), Lorenzo Valentini (CERN)
- 464. [Energy-aware compute resource modulation at the WLCG PIC Tier-1 site: drainage strategies, CPU frequency scaling, and predictive control](#), Jose Flix Molina (CIEMAT)
- 346. [Enhanced Data Integrity for Reliable WLCG Third-Party Copy Transfers](#), Hugo Gonzalez Labrador (CERN)
- 445. [Investigating Routing Anomalies and Performance Degradation in WLCG Networks \(Case Studies\)](#), Petya Vasileva (Michigan)
- 399. [Making WLCG Networks Visible: An Alarm and Visualization Platform](#), Petya Vasileva (Michigan)
- 269. [Replacing the Legacy Tier-0 Accounting System with Standard Technologies for WLCG Accounting](#), Ben Jones (CERN)
- 284. [Next-Generation Accounting Architecture for WLCG and EGI](#), Panos Paparrigopoulos (CERN)

- 552. [Deployment of site-focused security event detection capabilities](#), David Crooks (UKRI STFC)
- 313. [Enhancing High Availability and Disaster Recovery for Kubernetes Workloads at CERN](#), Jack Charlie Munday, Ricardo Rocha (CERN)
- 314. [Strengthening Vulnerability and SBOM Management in the CERN Container Registry](#), Jack Charlie Munday, Ricardo Rocha (CERN)
- 367. [Data-Driven Validation of HS23 with CMS Grid Job Monitoring Data](#), Robin Hofsaess
- 666. [A Blueprint for Emerging Centers: The NNU HEP Farm's Evolution from Local Cluster to Grid Integration](#), Kai Yi (Nanjing Normal Univ. (CN))
- 337. [Archiving 60 PB/month to tape — lessons learned and a look forward to Run-4](#), Julien Leduc (CERN)
- 612. [Operational Evolution of FTS3: A DevOps Driven Approach to elastic operations](#), Eric Vaandering (FNAL)
- 270. [Evaluating the scalability of CERN's HTCondor batch system towards the High-Luminosity LHC](#), Antonio Delgado Peris (CERN)
- 521. [CMS transfer rate estimates in Run-4](#), Katy Ellis (Science and Technology Facilities Council STFC (GB))
- 346. [Enhanced Data Integrity for Reliable WLCG Third-Party Copy Transfers](#), Hugo Gonzalez Labrador (CERN)

Apologies if we missed any!

Chapter 2: Facility Evolution Milestones



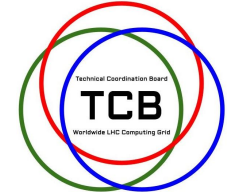
Some facility-driven milestones that came out of the HEPiX Spring 2026 Workshop:

Milestone	Description	Target Date
FE-1	With HEPiX, Survey on Operational Effort at Sites (Technical Sustainability), including AI for operations.	Q3 2026 (HEPiX Fall 2026 WS)
FE-2	Mini-challenges on tape performance, reported by each Tier-1 site, with WLCG DOMA.	Q4 2026
FE-3	Each Tier-1 site make a list of follow up actions (lessons learned) from DC27 (Feb. 2027) for the final report.	Q2 2027
FE-4	Studies (mini-challenges) on storage capacity vs. throughput, reported by each Tier-1 site and CERN/Tier-0.	TBD

← DC27

Risks include insufficient network or data transfer capacity to support new deployment strategies, scaling limitations (e.g., of data centers, pricing risks), lack of operational effort, etc.

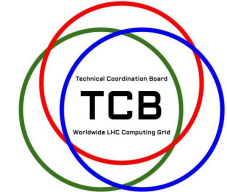
Chapter 3: Data Management - DOMA



- Gaps and blockers identified during DC24 and beyond:
 - Token refresh failures, database overloads in FTS3 and Rucio, site-specific monitoring, local infrastructure limitations such as storage and network saturation, market volatility for hardware procurements, IPv6 adoption, network traffic identification, jumbo frames identification, file sizes
 - Being actively addressed through ongoing R&D, mini-challenges, technology evaluations, monitoring enhancements, and infrastructure upgrades to ensure readiness for the HL-LHC era:
 - Examples: development of FTS4 (see [this slide](#)), Jumbo frames, New XRootD monitoring ([later slide](#)), and more.
- Next steps:
 - Continue with targeted mini-data challenges to test capacities and capabilities e.g., of FTS4.
 - **Data Challenge 2027 (DC27)**
 - Scheduled dates: Feb. 23 - Mar. 6, 2027 (two weeks)
 - Target: 50% HL-LHC (if feasible given current hardware)
 - Sites should not buy new hardware “just to pass DC27”.

This is another area where we have a concrete plan and well-agreed-upon milestones (see a [next slide](#) for details).

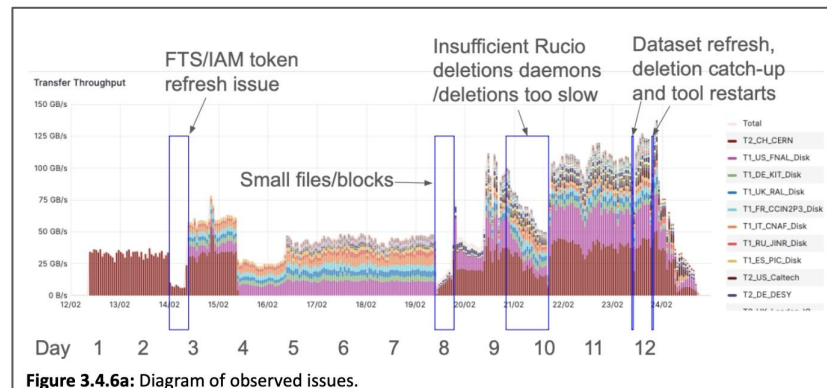
Chapter 3: Major DOMA Milestones



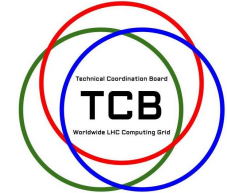
Milestone	Description	Target Date
DOMA-1	Define detailed DC27 targets (at the WLCG/HSF Workshop)	Q4 2026
DOMA-2	Data Challenge 2027 (DC27) - <i>50%-level HL-LHC</i>	Q1 2027
DOMA-3	Data Challenge 2029 (DC29) - <i>100%-level HL-LHC</i>	Q1 2029

See also [earlier DOMA-related milestones](#) in Chapter 1: Facility Evolution

[DC24](#) taught us that pre- and mini-challenges are key to success!



Chapter 3: FTS Milestones



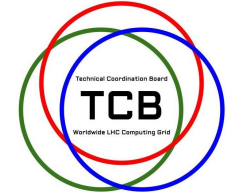
Timelines for the FTS4 roll-out.

Milestone	Description	Target Date
FTS-1	Scale testing with experiments (continuous)	H1+H2 2026
FTS-2	FTS4 in production for ATLAS (at CERN)	Q4 2026
FTS-3	FTS4 internal release for DC27	Q4 2026
FTS-4	FTS4 deployment at Tier-1 sites with full support	Q3 2027
FTS-5	FTS4 mini-challenges at Tier-1 sites	TBD



DC27

Chapter 3: XRootD Development Milestones



The XRootD team presented their updated plans at [OTF#9](#) in March 2026.

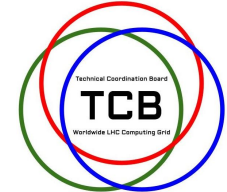
Milestone	Description	Target Date
XRD-1	XRootD data transfer performance tests (with DOMA) e.g., optimising file sizes	Q3 2026
XRD-2	Cloud-native XRootD	Q3 2026
XRD-3	XRootD monitoring of success/failures for operations	Q4 2026
XRD-4	XRootD documentation improvements	Q4 2027
XRD-5	XCACHE performance tests (related to IRIS-HEP AF GCs)	TBD
XRD-6	XRootD Token Challenge	TBD

← DC27

XRootD is an important project for the HEP community.

N.B. XRootD Monitoring (as a service) is discussed in [Chapter 6 - Services](#).

Chapter 5: Workflow Management



Evolution will be a truly “bottom-up” process in each experiment and development platform, but there are several areas for strategic cooperation and coordination:

Capability-Aware Scheduling

- Hardware-aware matchmaking
- Hybrid CPU/GPU workflows
- Whole-node and accelerator-aware execution

Fluid Resource Integration

- HPC and cloud allocations
- Opportunistic capacity
- Dynamic workload buffering

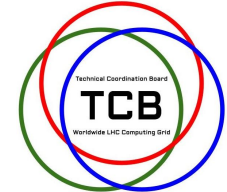
Emerging Workflow Models

- ML training and inference
- Interactive analysis facilities
- Flexible workflow orchestration

Cross-Experiment Coordination

- Shared provisioning concepts
- Common resource catalogues
- Coordination and standardisation

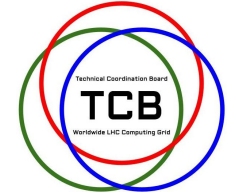
Chapter 6: Services - Accounting Milestones



Timelines mostly driven by end-of-life of legacy systems EGI Accounting Portal, APEL.

Milestone	Description	Target Date
ACC-1	Assessment of AUDITOR and deployment by the pioneer sites. (Underway)	Q4 2026
ACC-2	Eliminate the dependency on the EGI Accounting Portal. Deploy the initial implementation of the new repository and portal. (Underway)	Q4 2026
ACC-3	Remove the dependency on the APEL legacy system. This requires a new central component to consume data directly from the queue containing the raw data published by sites.	Q3 2027
ACC-4	Mass deployment campaign of AUDITOR at the European WLCG sites (Start: Q1 2027)	Q4 2028

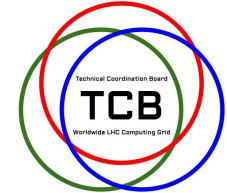
Chapter 6: Services - CRIC Milestones



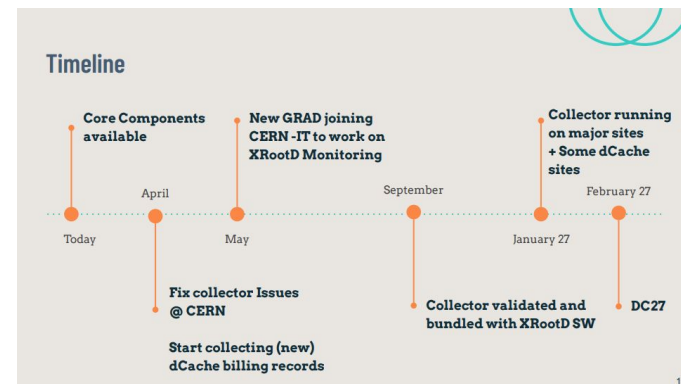
Milestone	Description	Target Date
CRIC-1	CRIC MCP server prototype: define structured interface for AI agent access to CRIC resource information.	Q2 2027
CRIC-2	GPU topology recorded in CRIC	Q2 2027**
CRIC-3	Records SRRs for Storage Space Accounting in WLCG CRIC	Q2 2027
CRIC-4	CMS CRIC Python 3 migration	Q3 2027
CRIC-5	Cleanup & audit: systematic review of stale services, endpoints and user records; establish automated cleanup mechanisms.	Q3 2027

*** Depends on the evolution of the community's needs.*

Chapter 6: Services - XRootD Monitoring



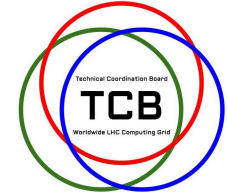
- XRootD Monitoring is important for both WLCG and CMS
 - For WLCG: XRootD represents a large and growing fraction of data transfers, yet it remains largely unmonitored. Closing this visibility gap is essential for operations, capacity planning, and network usage validation (all critical as we approach HL-LHC).
 - For CMS: XRootD monitoring also enables data popularity tracking and and failure monitoring.
- Two use cases, two implementations:
 - Native XRootD servers — monitored via the OSG Go-based collector
 - dCache servers exposing an XRootD door
- Both implementations exist and are actively being pursued. All necessary components are in place, but significant effort remains for validation and broad deployment across WLCG sites.
- See related CHEP [presentation](#).



P. Paparrigopoulos [presenting](#) at the CMS O&C Week Open Session, March 2026.

Details of the plan in [the next slide](#).

Chapter 6: Services - XRootD Monitoring Milestones

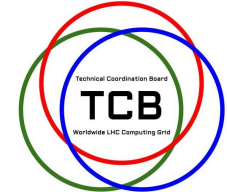


Milestone	Description	Target Date
MONIT-1	Deployment and validation of Go-based collector at CERN	Q3 2026
MONIT-2	Bundling of the solution with XRootD	Q4 2026
MONIT-3	Validation of the dCache solution (Kafka bookkeeping path).	Q4 2026
MONIT-4	Deployment of the monitoring solution to a few major sites	Q1 2027
MONIT-5	Validation of the new dCache v11.2 solution (Direct messaging without local Kafka)	Q2 2027
MONIT-6	Failure monitoring (bound to XRootD plans)	Q1 2028
MONIT-7	Completed XRootD monitoring deployment across WLCG	Run-4 start

← DC27

← DC29

Chapter 7: Token Transition Milestones

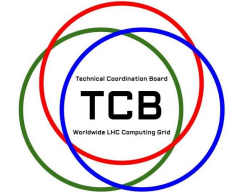


Milestone	Description	Target Date
TOK-1	Risk assessment including the effects of outages	Q2 2026
TOK-2	First usage of tokens for tapes	Q3 2026
TOK-3	First usage of tokens in ATLAS jobs	Q3 2026
TOK-4	Grid Job Challenge – stress WM fwks & token issuers	Q4 2026
TOK-5	CMS production jobs – some fraction using only tokens	Q3 2027
TOK-6	Completion of the X509 / VOMS phaseout	H2 2028

← DC27

- Preparing for **DC27** is an important intervening milestone.
- Milestones for user jobs have not been agreed upon yet.
- While a lot work remains to be done, we foresee **significant progress** year by year, thanks to the **efficient collaboration** between all parties concerned !
- See related CHEP26 presentations on [AAI at CERN](#), [Token Transition WG](#).

Chapter 7: Security Milestones

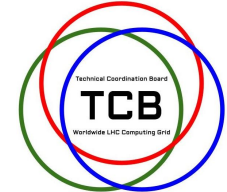


Security milestones are all tentative and need to be discussed over the next few months.

Policy Milestones:

Milestone	Description	Target Date
SEC-1	Identify outdated policies and plan discussions with stakeholders	Q2 2026
SEC-2	Harmonization of policy repository and update workflow	Q3 2026
SEC-3	Proposal for MoU adaptation (strengthen security posture)	Q4 2026
SEC-4	Roadmap for policy review	Q1 2027
SEC-5	Self-assessment checklist (as maturity indicator)	Q3 2027

Chapter 7: Security Milestones



Containment Workflows:

Milestone	Description	Target Date
SEC-6	Validate existing containment workflows in case of compromise	H1 2026
SEC-7	Recommendations/playbook	H2 2026

Security Skills / Improvement Gaps:

Milestone	Description	Target Date
SEC-8	Forensics course started	Q2 2025
SEC-9	Skill gaps survey	H2 2026
SEC-10	Security training roadmap	H1 2027